

Xanthomonas Transformation Protocol

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To introduce plasmid DNA into *Xanthomonas* via electroporation for gene knockout, complementation, or plasmid expression studies.

Materials

- Xcc strain
- NB or NYG broth
- Selective agar plates
- SOC recovery medium
- Cold sterile distilled water
- Cold 10% glycerol
- Plasmid DNA
- Ice bath
- Refrigerated centrifuge (4°C)
- 0.1 cm electroporation cuvettes (pre-chilled)
- Electroporator (e.g., Eppendorf Multiporator)
- 28°C shaking incubator (200 rpm)

Procedure

Starter Culture

1. Pick isolated colony.
2. Inoculate into 10 mL broth.
3. Grow 48 h at 28°C.

Expand to Log Phase

1. Transfer 5 mL starter culture into 500 mL broth.
2. Incubate at 28°C with shaking.
3. Grow to A600 = 0.6–0.8 (mid-log phase).

Chill Cells

1. Place flask on ice for 15 min.
2. Centrifuge at 6000 × g for 10 min at 4°C.
3. Discard supernatant.

Washing

Purpose: Remove salts to prevent electrical arcing.

1. After centrifugation, carefully decant or pipette off the supernatant without disturbing the pellet.
2. Add cold sterile distilled water to fully resuspend the pellet (typically equal to the original culture volume or enough to loosen the pellet completely).
3. Gently resuspend the cells by pipetting up and down or flicking the tube until no clumps remain.
4. Centrifuge again at 6000 × g for 10 min at 4°C.
5. Carefully remove and discard the supernatant.

☒ Explore the Notebook  once more (total of two washes).
glycerol to resuspend the pellet.

8. Gently resuspend until homogeneous.
9. Centrifuge at $6000 \times g$ for 10 min at 4°C .
10. Carefully remove and discard the supernatant.
11. Repeat procedure 7 - 10 once more (total of two washes).
12. Resuspend final pellet in 1 mL cold 10% glycerol.

Prepare Competent Cells

1. Measure OD and estimate cell density.
2. Aliquot 50 μL portions ($\sim 10^8$ – 10^9 cells).
3. Use immediately or store at -80°C .

Electroporation

1. Add plasmid DNA (typically 30 ng) to 50 μL cells.
2. Transfer to chilled 0.1 cm cuvette.
3. Apply single pulse:
 - Field strength: 14 kV/cm (optimized)
 - Capacitance: 10 μF
 - Resistance: 56 Ω serial / 600 Ω parallel
4. Immediately add 1 mL SOC medium after pulse.

Recovery

1. Transfer to culture tube.
2. Incubate at 28°C , 200 rpm for 2–4 h.
(Allows antibiotic resistance expression.)

Selection

1. Plate 100 μL on selective agar plates.
2. Incubate at 28°C for ~ 3 days.

Calculate Transformation Efficiency

Transformation efficiency = Number of transformants / μg plasmid DNA²